

Week 1 Tutorial: Python Foundations

ILO

- **KU1:** Grasp the relation between source code, the interpreter, and the machine.
- **KU3:** Design algorithms to solve simple programming problems.
- **CS1:** Structure small programs by the use of concepts such as iterations, functions, modules, classes, and methods.
- **CS3:** Form readable, descriptive and well-documented program code.
- **CS6:** Build basic interactive programs with text-based (and graphical) user interfaces.
- **CS8:** Use programming tools such as text editor, command line interface, and IDE (integrated development environment).

Objectives

After completing this tutorial, you should be able to:

1. Explain that Python source code is run by an interpreter, one instruction at a time.
2. Create and run a Python program that displays text using `print()`.
3. Store values in variables and use them in expressions and output.
4. Read text and integer input from a user using `input()` and `int()`.
5. Use an `if/else` statement to select between two actions.
6. Use a `for` loop and `range()` to repeat an instruction a fixed number of times.
7. Predict, run, and modify short Python programs to check your understanding.

Lesson Plan

We start with the idea that programming is giving precise instructions to a computer. We code together to get a few tiny programs running immediately. During this coding activity, we discuss as only that theory which is relevant and needed, avoiding overburdening ourselves with loads of theoretical aspects.

A useful first lesson sequence:

1. What a program does, control flow

- Show:

```
print("Hello, world!")
```

- Change the text together and introduce multiple messages. Lets see that code is read top to bottom in a sequential way.

2. Variables

- Present variables as named containers for values:

```
name = "Ada"
age = 18
print(name)
print(age)
```

- what do we predict the output to be (before running the program)?

3. Input and output

- Make programs personal and interactive:

```
name = input("What is your name? ")
print("Hello,", name)
```

- Then type conversion:

```
age = int(input("How old are you? "))
print("Next year you will be", age + 1)
```

4. Conditions

- We discuss decisions through familiar rules:

```
temperature = int(input("Temperature: "))

if temperature < 0:
    print("Wear a coat.")
else:
    print("Normal weather.")
```

5. Loops

- Start with repetition that are simply displayed, so that we can see:

```
for number in range(1, 6):
    print(number)
```

- Then we make a simple task, such as adding five entered numbers.